Linear Functions

Definition

A **linear function** has the form

$$f(x) = mx + b,$$

where m is the slope and $(0, b)$ is the y-intercept. Its graph is a straight line and its domain is $(-\infty, \infty)$.

Key features

- constant rate of change m;
- $m > 0$: increasing;
- $m < 0$: decreasing;
- $m = 0$: constant;
- zeros occur where $f(x) = 0$.

• Zeros of the function is x-intercept.
Or zeros of function is what x-value let $y=0$

First example

$$f(x) = -3x + 6$$

Identify the slope, y-intercept, direction, and zero.

• $m = -3$

• y-intercept: $(0, 6)$

• decreasing

• $-3x + 6 = 0$

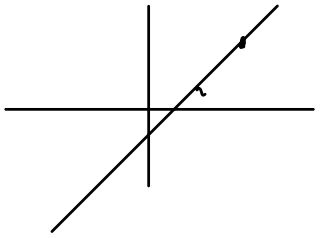
$x = 2$

$(2, 0)$

Slope and Rate of Change

For two distinct points (x_1, y_1) and (x_2, y_2),

$$\boxed{m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x}} \qquad (x_1 \neq x_2).$$



For a linear function, this value is the same between *every* pair of points.

Example. Find the slope of the line through $(-2, 5)$ and $(4, -1)$. Then interpret its sign.

- slope tells us how much y changes when x changes.
- Δy = change in y Δx = chang in x.
- m = chang in output divide by change in input.

$$m = \frac{\Delta y}{\Delta x} = \frac{-1-5}{4-(-2)} = -6$$

$(-2, 5)$ $(4, -1)$

x_1 y_1 x_2 y_2

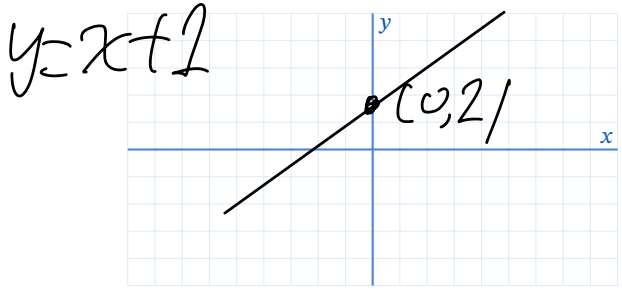
$$\frac{-1-5}{-2-4} = \frac{y_2 - y_1}{x_1 - x_2}$$

$$\frac{y_2 - y_1}{x_2 - x_1} \qquad \frac{y_1 - y_2}{x_1 - x_2}$$

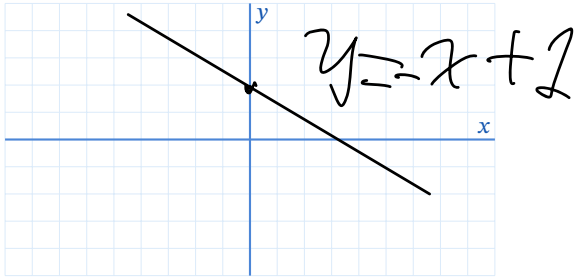
Read the Sign of the Slope

For each graph, decide whether the slope is positive, negative, or zero. Then sketch one possible line.

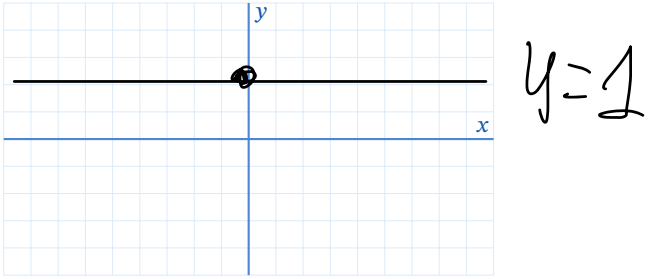
1. Increasing



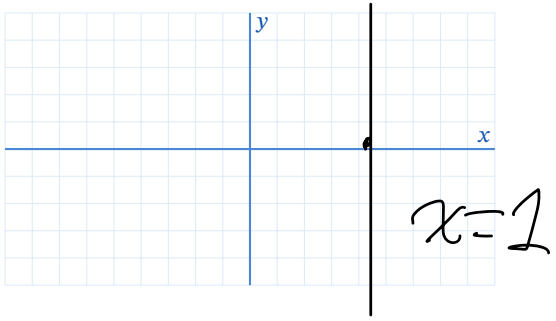
2. Decreasing



3. Constant



4. Vertical line: is it a function?



- def:
  input a x value, it will output a unique y value.

Linear or Nonlinear from a Table

Compute the change in y for each equal change in x. A constant rate of change signals a linear function.

Table A

x	-1	0	1	2
$f(x)$	0	4	8	12

Linear or nonlinear? If linear, find m.

$f(x) = mx + b$,

- $\Delta y, \Delta x$ must be the same for every pair of point.
- $x_1 = -1, x_2 = 0$, $\Delta x = 0 - (-1) = 1$
  $y_1 = 0, y_2 = 4$, $\Delta y = 4 - 0 = 4$
- $x_1 = 0, x_2 = 1$, $\Delta x = 1 - 0 = 1$
  $y_1 = 4, y_2 = 8$ $\Delta y = 8 - 4 = 4$
- $x_1 = 1, x_2 = 2$ $\Delta x = 1$
  $y_1 = 8, y_2 = 12$, $\Delta y = 4$

so the function is linear.

$\frac{\Delta y}{\Delta x} = \frac{4}{1} = 4$ $y = 4x + b$ $\Rightarrow$ $y = 4x + 4$

$4 = 4 \cdot 0 + b$, $b = 4$

Table B

x	-2	-1	0	1	2
$y = g(x)$	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4

Linear or nonlinear? Explain using rates of change.

$\Delta x = 1$

$-1 - (-2) = 1$, $0 - (-1) = 1 \cdots$

$\frac{1}{2} - \frac{1}{4} = \frac{1}{4}$ "Δy_1" $= \frac{1}{4}$

$1 - \frac{1}{2} = \frac{1}{2}$ "Δy_2" $= \frac{1}{2}$

$2 - 1 = 1$

$\vdots$

so is not linear.

Write a Linear Equation

Choose a form that matches the information given.

if partical information are given, how to find linear function.

Information	Useful form	What to do
Slope and y-intercept	$y = mx + b$	Substitute m and b.
Slope and one point	$y - y_1 = m(x - x_1)$	Use point–slope form, then simplify if needed.
Two points	$m = \frac{y_2 - y_1}{x_2 - x_1}$	Find m first, then use either point.

Example. Find the equation of the line through $(-1, 2)$ and $(3, 10)$.

1). given $m=3, b=4. \Rightarrow y=3x+b.$

2). given $m=2$, $(1,3)$ is on the graph.

$y-y_1=m(x-x_1)$

$y-3=2(x-1)$

$y-3=2x-2$

$y=2x+1$

3) $m=\frac{10-2}{3-(-1)}=\frac{8}{4}=2$

$y=2x+b$

$10=2\cdot 3+b$

$10-6=b$

$b=4$

$y=2x+4$

Zeros and Intersections

Zero of a linear function

To find the zero of f, solve

$$f(x) = 0.$$

The corresponding graph point is an x-intercept $(x, 0)$.
Example: $f(x) = 4x - 1$.

$4x - 1 = 0$

$x = \frac{1}{4}$

$\left(\frac{1}{4}, 0\right)$

Intersection of two functions

To find where f and g meet, solve

$$f(x) = g(x),$$

then substitute the x-value to find y.
Example: $f(x) = 4x - 1$ and $g(x) = -2x + 5$.

find intersection of $f(x)$ and $g(x)$

$f(x) = g(x)$

$4x - 1 = -2x + 5$

$6x = 6$

$x = 1$

$f(1) = 4 \cdot 2 - 1 = 3$

$g(1) = -2 \cdot 1 + 5 = 3$

$(1, 3)$ is the intersection.

Linear Inequalities

Solve an inequality using the same algebra as an equation, with one crucial rule:

> Multiplying or dividing by a negative number reverses the inequality sign.

1. If $f(x) = 4x - 1$, solve $f(x) > 0$.
2. If $f(x) = 4x - 1$ and $g(x) = -2x + 5$, solve $f(x) \leq g(x)$.

1). $4x-1>0$

$4x>1$

$x>\frac{1}{4}$

So when $x>\frac{1}{4}$, $f(x)>0$.

2). $4x-1\leq -2x+5$

$6x\leq 6$

$x\leq 1$

So when $x\leq 1$, $f(x)\leq g(x)$.

Linear Model Practice

A gym charges a $25 registration fee and $18 per month.

Total cost = starting fee + monthly cost × month.

(a) Write the total cost $C(t)$ after t months.

(b) Identify and interpret the slope and y-intercept.

(c) Find $C(8)$ and explain its meaning.

(d) A second gym charges $C_2(t) = 22t + 5$. When do the costs match?

(e) Which plan is less expensive after one year?

(a) $C(t) = 25 + 18 \cdot t = 18t + 25$

(b) $m = 18$, $(0, 25)$

(c) $C(8) = 18 \cdot 8 + 25 = 169$, total cost up to 8 month.

(d) $22t + 5 = 18t + 25$

$4t = 20$

$t = 5$

$(5, 115)$

$C(5) = 18 \cdot 5 + 25$

$= 115$

(e) $C(12) = 241$ ✓

$C_2(12) = 269$